

docs_chain — HelixConstitution Integration

Revision: 1 **Last modified:** 2026-05-29T00:00:00Z **Status:** Design documentation — DESCRIBES the integration. The integration itself is PLANNED (Phase 6) and OPERATOR-GATED. This document does NOT implement it. **Authority:** Operator mandate 2026-05-29 (docs_chain initiative) **Design provenance:** authoritative Phase-0 DESIGN / RESEARCH / PLAN live in the consuming project research tree (docs/research/docs_chain/); this document is the self-contained specification.

Status and scope boundary

PLANNED (Phase 6) — OPERATOR-GATED. Phase 6 of PLAN.md covers remote-repo creation, the constitution-submodule pointer add, and the governance-file edits (Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md). Those steps are performed by the operator, not by an agent (§11.4.66 / §11.4.35 / §11.4.26).

This document **describes** the integration model so the operator and future consuming projects know the plan. It is explicitly **not** an implementation: nothing here edits any governance file, creates any remote, or adds any submodule. The reader should treat every "the constitution submodule will ..." statement as the DESIGNED Phase 6 outcome, not a current fact.

1. The distribution model

docs_chain ships as its own vasic-digital submodule and is consumed as a CORE part of the HelixConstitution submodule. Once Phase 6 lands, any project that already inherits the constitution submodule gets docs_chain **by reference** — the same inheritance mechanism the project already uses for the constitution's CLAUDE.md / AGENTS.md / Constitution.md.

graph TD

```
    DCRepo[vasic-digital/docs_chain<br/>standalone submodule] -->
    CONST[HelixConstitution submodule]
```

```
CONST --> P1[Consuming project A]
CONST --> P2[Consuming project B]
CONST --> P3[Consuming project N]
P1 -.registers.-> C1[.docs_chain/contexts/*.yaml]
P2 -.registers.-> C2[.docs_chain/contexts/*.yaml]
P3 -.registers.-> C3[.docs_chain/contexts/*.yaml]
```

The engine (Go binary + builtins) is shared and project-agnostic; the **contexts** (`.docs_chain/contexts/*.yaml`) are per-project data the consumer owns and registers. This mirrors the §11.4.28 decoupling rule: the submodule never carries project-specific context; the consumer injects its chains via config.

2. Inheritance model

Per §11.4.35 (canonical-root inheritance clarity), there are two layers:

- **Canonical root (the constitution submodule).** Hosts the `docs_chain` engine reference + the universal anchor describing the mandate `docs_chain` satisfies. Project-agnostic.
- **Consumer (this and every other project).** Registers its own `.docs_chain/contexts/*.yaml`, invokes the engine via the path the constitution exposes, and lists its chains in its own docs.

A consuming project does NOT copy the engine; it references it at the path the constitution exposes (the same pattern §11.4.80 uses for the `codegraph_*` scripts — inherited by reference, never copied).

3. Config-discovery convention

Status: PLANNED (Phase 4 loader + Phase 6 distribution).

The discovery convention a consuming project follows out of the box:

1. Engine resolves contexts from `<project-root>/docs_chain/contexts/`.
2. Each `*.yaml` is one independent context (ARCHITECTURE §9).
3. `state.json` lives at `<project-root>/docs_chain/state.json`, gitignored, regenerated on demand (§11.4.77 — the regeneration mechanism is `docs_chain sync`).
4. `docs_chain sync --all / docs_chain verify --all` walk every context file; per-context commands take the context name.

No hardcoded project paths live in the engine — every path is supplied by the consumer's YAML (§11.4.28-B decoupling).

4. §11.4.x anchors docs_chain satisfies / supersedes

docs_chain is the mechanical engine behind a family of existing documentation-sync mandates. Once a consuming project registers the matching context (see [USE_CASE_CATALOGUE.md](#)), the engine enforces the anchor in place of the ad-hoc script.

Anchor	What it mandates	docs_chain mechanism	Recipe
§11.4.12	Issues_Summary always-sync	derive summary ← markdown + early-cutoff	(a)
§11.4.53	Fixed_Summary parity	derive fixed_summary ← Fixed.md	(b)
§11.4.45	Status.md maintenance	status-doc context	(c)
§11.4.56	Status_Summary two-audience parity	derive status_summary ← Status.md	(c)
§11.4.57	README doc-link section	multi-input derive into README	(f)
§11.4.59	README always-sync export	derive README .html/.pdf	(f)
§11.4.60	Documentation composite-sync	the composite of all derive edges	(a)–(h)
§11.4.65	Universal markdown export	derive .html + .pdf per .md	(e), (g)
§11.4.86	Roster/corpus auto-sync (fingerprint not mtime)	members-fingerprint node + content-hash detection	(d)
§11.4.93	Workable-items DB single source of truth	bidirectional sync edge md ↔ sqlite	(a)
§11.4.95	DB tracked + WAL-checkpoint commit	atomic SQLite-txn commit phase	(a)
§12.10	CONTINUATION maintenance	derive CONTINUATION exports	(h)
§11.4.44	Document revision header		(a)–(h)

Anchor	What it mandates	docs_chain mechanism	Recipe
		source-author owns header; engine keeps exports in sync	
§9.2	Zero-risk data safety	atomic-rename commit + rollback + optional hardlink backup	engine-wide
§11.4.6	No-guessing	both-dirty sync → conflict (exit 2), never silent merge	engine-wide
§11.4.50	Deterministic consistency	verify is the deterministic sink-side check; byte-stable transforms	engine-wide
§11.4.69	Sink-side positive-evidence	per-run evidence to qa-results/docs_chain/<run-id>/	engine-wide

docs_chain does NOT replace the authoring discipline of any anchor (e.g. the operator still writes the §11.4.44 revision header into the source); it replaces the **mechanical sync** the anchor requires.

5. What a consuming project gets for free vs must register

5.1 For free (provided by the constitution submodule once Phase 6 lands)

- The docs_chain engine (Go binary + builtins: pandoc-html, weasyprint-pdf, colorize-html, members-fingerprint, gen-summary, md-to-sqlite / sqlite-to-md adapters).
- The CLI (sync / verify / watch / graph / doctor) and exit-code contract.
- Atomic-commit + rollback + conflict semantics (ARCHITECTURE §5, §8).
- Content-hash change detection (§11.4.86 not-mtime).
- The use-case recipe catalogue (this submodule's USE_CASE_CATALOGUE.md) as a starting point.

5.2 Must register (the consumer owns)

- Its own .docs_chain/contexts/*.yaml describing its chains.

- The exec: transform scripts its contexts reference (during migration — its existing generators).
 - The .gitignore entries for state.json and *.docs_chain.tmp.
 - The CI / pre-build wiring that calls docs_chain verify (operator-gated where it edits a gate file — Phase 7.4).
 - Any project-specific node kinds beyond the universal set (rare).
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6. Out-of-the-box adoption checklist

Status: PLANNED — usable once Phase 6 distribution + Phase 4 CLI land.

A project follows these steps to start using docs_chain:

1. Inherit the constitution submodule (already done by every governed project).
 2. Build / obtain the docs_chain binary (ARCHITECTURE §11; USER_GUIDE §3).
 3. mkdir -p .docs_chain/contexts and add state.json + *.docs_chain.tmp to .gitignore.
 4. Copy the relevant recipes from USE_CASE_CATALOGUE.md, adjusting paths.
 5. docs_chain doctor to validate every context.
 6. docs_chain sync --all to bring everything in sync; inspect any conflicts (USER_GUIDE §9).
 7. Wire docs_chain verify --all into CI / the pre-build gate (operator-gated where it edits a gate file).
 8. Optionally run docs_chain watch during interactive development.
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7. Phase-6 governance integration (operator-gated — described, not done)

Per PLAN.md Phase 6, the operator (NOT an agent) performs:

- (6.1) create vasic-digital/docs_chain on GitHub + GitLab via gh / glab (§11.4.66 — agent must not create remotes).
- (6.2) install_upstreams recipes + multi-mirror push (§11.4.36 / §2.1).
- (6.3) add the docs_chain submodule to the constitution submodule (operator authorises the governance edit).
- (6.4) governance-file integration — the new anchor + propagation gate in Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md (§11.4.35 / §11.4.26 — operator edits governance).

- (6.5) `helix-deps.yaml` manifest (§11.4.31), plus the cross-project bootstrap Challenge that proves a fresh consumer can register a chain and `sync` works with captured wire-evidence.

This documentation set is the agent-executable companion to those operator-gated steps; it carries no governance edits itself.